Launchpad-Mixxx-DJ

Play a full DJ set in Mixxx with one Novation Launchpad MK2

User Manual | Meta Hackathon | Mixxx 2.5.x | macOS

Covers: installation, Mixxx setup, emulator tour, both pad maps, your first transition, hardware mode, troubleshooting.

1. What this is

Launchpad-Mixxx-DJ turns a single Novation Launchpad MK2 into two CDJ-style decks plus a mixer and FX unit for Mixxx (2.5.x) on macOS. Mixxx stays in charge of all audio; the app translates pad presses into Mixxx commands over MIDI and drives the pad LEDs. No DJ controller hardware is required: the app ships an on-screen emulator that drives Mixxx while it is open.

- Two profiles: User 2 = decks (Deck A left, Deck B right). User 1 = mixer + FX.
- Green = increase, red = decrease (hold: 1% per 20 ms). Yellow = one action per press.
- Knobs use 3 pads (down / reset-to-centre / up); faders use 2 pads (down / up).
- Held pads glow bright; released pads glow dim in their colour.

2. Installation

Requirements

- macOS with Python 3.11+ (tkinter included)
- Mixxx 2.5.x
- Novation Launchpad MK2 only if using hardware mode

Steps

1. Unzip the release and open a Terminal in the project folder.
2. Create a virtual environment: python3 -m venv .venv && source .venv/bin/activate
3. Install dependencies: pip install -r requirements.txt
4. Sanity check: python3 -m pytest tests/ -q (expect 16 passed).

3. Connecting Mixxx (one time)

1. Copy mixxx_mapping/LaunchpadMK2Bridge.midi.xml into Mixxx's user controllers folder. On macOS (Mixxx 2.5): ~/Library/Containers/org.mixxx.mixxx/Data/Library/Application Support/Mixxx/controllers/ (create it if missing).
2. Start the emulator (section 4) so Mixxx can see the virtual port.
3. In Mixxx: Preferences > Controllers > select the LaunchpadMK2Bridge device.
4. Enable it and choose the 'Launchpad MK2 DJ Bridge' mapping, then OK.
5. If the device is missing, close and reopen Preferences while the emulator runs.

Technical note: the mapping is declarative XML (notes for buttons, CCs for continuous controls). The app owns all repeat and profile logic; Mixxx only receives commands and plays audio.

4. Emulator tour

Launch it with: PYTHONPATH=src python3 main.py --emulator

- Top row: Session / User 1 / User 2 / Mixer. User 1 selects mixer+FX, User 2 selects decks. The active one glows.
- Main 8x8 grid: click-and-hold pads exactly like hardware. Row 0 is the top row.
- Profile label under the top row tells you which layout is active.
- Right panel: connection status plus a live log of every command sent to Mixxx, e.g. [Channel1] play 1.00 or [Master] crossfader 0.57.
- MIDI live means the CoreMIDI virtual port is open and Mixxx can hear it.

Clicking technique

- Buttons: single click = one action. Holding does nothing extra; release re-arms.
- Knobs/faders: press AND HOLD the green (up) or red (down) pad; value moves 1% per 20 ms.
- Yellow pad inside a knob = snap that control back to centre (50%).
- Switching profiles or releasing cancels any in-progress hold.

5. Decks profile (User 2)

Deck A owns columns 0-3 ([Channel1]), Deck B mirrors it on columns 4-7 ([Channel2]). Bottom row holds shared crossfader and headphone controls.

Pad	Control	Type	Action	LED	Mixxx target
R0C0	deckA.load	button	press once	yellow	[Channel1] LoadSelectedTrack
R0C1	deckA.play	button	press once	yellow	[Channel1] play
R0C2	deckA.cue	button	press once	yellow	[Channel1] cue_default
R0C3	deckA.sync	button	press once	yellow	[Channel1] beatsync
R0C4	deckB.load	button	press once	yellow	[Channel2] LoadSelectedTrack
R0C5	deckB.play	button	press once	yellow	[Channel2] play
R0C6	deckB.cue	button	press once	yellow	[Channel2] cue_default
R0C7	deckB.sync	button	press once	yellow	[Channel2] beatsync
R1C0	deckA.gain	knob	hold: -1%/20ms	red	[Channel1] pregain
R1C1	deckA.gain	knob	reset 50%	yellow	[Channel1] pregain
R1C2	deckA.gain	knob	hold: +1%/20ms	green	[Channel1] pregain
R1C4	deckB.gain	knob	hold: -1%/20ms	red	[Channel2] pregain
R1C5	deckB.gain	knob	reset 50%	yellow	[Channel2] pregain
R1C6	deckB.gain	knob	hold: +1%/20ms	green	[Channel2] pregain
R2C0	deckA.rate	knob	hold: -1%/20ms	red	[Channel1] rate
R2C1	deckA.rate	knob	reset 50%	yellow	[Channel1] rate
R2C2	deckA.rate	knob	hold: +1%/20ms	green	[Channel1] rate
R2C4	deckB.rate	knob	hold: -1%/20ms	red	[Channel2] rate
R2C5	deckB.rate	knob	reset 50%	yellow	[Channel2] rate

R2C6	deckB.rate	knob	hold: +1%/20ms	green	[Channel2] rate
R3C0	deckA.filter	knob	hold: -1%/20ms	red	[Channel1] filterHigh
R3C1	deckA.filter	knob	reset 50%	yellow	[Channel1] filterHigh
R3C2	deckA.filter	knob	hold: +1%/20ms	green	[Channel1] filterHigh
R3C4	deckB.filter	knob	hold: -1%/20ms	red	[Channel2] filterHigh
R3C5	deckB.filter	knob	reset 50%	yellow	[Channel2] filterHigh
R3C6	deckB.filter	knob	hold: +1%/20ms	green	[Channel2] filterHigh
R4C1	deckA.pfl	button	press once	yellow	[Channel1] pfl
R4C5	deckB.pfl	button	press once	yellow	[Channel2] pfl
R7C0	mixer.headmix	button	press once	yellow	[Master] headMix
R7C3	mixer.crossfader	fader	hold: -1%/20ms	red	[Master] crossfader
R7C4	mixer.crossfader	fader	hold: +1%/20ms	green	[Master] crossfader
R7C6	mixer.headcue	button	press once	yellow	[Master] headSplit

6. Mixer + FX profile (User 1)

Channel faders, mutes, cue buttons, master/headphone knobs, FX mix and parameter knobs, and FX routing buttons.

Pad	Control	Type	Action	LED	Mixxx target
R0C0	mixer.ch1	fader	hold: -1%/20ms	red	[Channel1] volume
R0C1	mixer.ch1	fader	hold: +1%/20ms	green	[Channel1] volume
R0C3	mixer.ch2	fader	hold: -1%/20ms	red	[Channel2] volume
R0C4	mixer.ch2	fader	hold: +1%/20ms	green	[Channel2] volume
R0C6	mixer.crossfader	fader	hold: -1%/20ms	red	[Master] crossfader
R0C7	mixer.crossfader	fader	hold: +1%/20ms	green	[Master] crossfader
R1C0	mixer.mute1	button	press once	yellow	[Channel1] mute
R1C1	mixer.mute2	button	press once	yellow	[Channel2] mute
R1C3	mixer.pfl1	button	press once	yellow	[Channel1] pfl
R1C4	mixer.pfl2	button	press once	yellow	[Channel2] pfl
R2C0	mixer.master	knob	hold: -1%/20ms	red	[Master] gain
R2C1	mixer.master	knob	reset 50%	yellow	[Master] gain
R2C2	mixer.master	knob	hold: +1%/20ms	green	[Master] gain
R2C4	mixer.headmix	knob	hold: -1%/20ms	red	[Master] headMix
R2C5	mixer.headmix	knob	reset 50%	yellow	[Master] headMix
R2C6	mixer.headmix	knob	hold: +1%/20ms	green	[Master] headMix
R4C0	fx.mix	knob	hold: -1%/20ms	red	[EffectRack1_EffectUnit1] mix
R4C1	fx.mix	knob	reset 50%	yellow	[EffectRack1_EffectUnit1] mix
R4C2	fx.mix	knob	hold: +1%/20ms	green	[EffectRack1_EffectUnit1] mix

R4C4	fx.param	knob	hold: -1%/20ms	red	[EffectRack1_EffectUnit1_Effect1] parameter1
R4C5	fx.param	knob	reset 50%	yellow	[EffectRack1_EffectUnit1_Effect1] parameter1
R4C6	fx.param	knob	hold: +1%/20ms	green	[EffectRack1_EffectUnit1_Effect1] parameter1
R6C0	fx.enable1	button	press once	yellow	[EffectRack1_EffectUnit1] enable
R6C1	fx.enableA	button	press once	yellow	[EffectRack1_EffectUnit1] group_[Channel1]_enal
R6C2	fx.enableB	button	press once	yellow	[EffectRack1_EffectUnit1] group_[Channel2]_enal

7. Your first transition (step by step)

1. Load two tracks into Mixxx's library (Auto DJ / crates).
2. User 2 profile. Deck A: click LOAD, then PLAY. Track A starts.
3. Deck B: click LOAD, then CUE, then SYNC to match tempo, then PLAY.
4. Hold the crossfader ^ pad (row 7) for ~2 seconds to sweep fully from A to B.
5. Press User 1. Click FX>B to route the effect to Deck B, then click FX to enable it.
6. Hold FX MIX ^ for ~1 second to bring the effect in. Adjust FX PRM to taste.
7. Press User 2 to return to the decks and continue mixing. Repeat for every transition.

Tip: short taps under 20 ms do nothing by design (below one repeat tick). Deliberate holds of 50-150 ms give fine 2-7% nudges.

8. Physical Launchpad mode

1. Quit the emulator. Plug in the Launchpad MK2 (Session layout).
2. Run: PYTHONPATH=src python3 main.py (hardware path; the physical
3. T01 spike still has to be run against a real MK2 - the emulator is
4. the supported route until then).
5. Physical User 1 / User 2 buttons switch profiles; the app redraws the LEDs.
6. Everything else behaves exactly like the emulator, including colours.

9. Troubleshooting

- Mixxx doesn't react: mapping not installed/enabled (section 3); Preferences opened before the emulator started.
- Two LaunchpadMK2Bridge ports listed: an old emulator instance is still running; quit it.
- Value jumps on first touch: by design the app owns 0-100% state and Mixxx follows (no soft-takeover in v1).
- LED colours look wrong: dim/bright palette needs calibration against a real MK2 (single spot in protocol.py).
- Emulator says MIDI unavailable: not on macOS or CoreMIDI denied; commands still logged.

10. Appendix

Timing contract

Knob/fader holds step on a 20 ms tick: 1% at 20 ms, 50% at 1 s, 100% at 2 s from 0%, clamped at both ends. Symmetric for decrease. If both direction pads are held, the last press wins; release cancels the hold. Buttons fire once per press.

Verification status

16/16 deterministic tests pass (fake clock and fake endpoints). Physical acceptance runs on a real MK2 with live Mixxx are recorded as Blocked in docs/VERIFICATION.md until hardware timing (scheduling jitter, LED palette) is measured.

Files

- src/launchpad_mixxx/: protocol, clock, mapping, engine, router, leds, endpoints, coremidi_bridge, app, emulator.
- mixxx_mapping/LaunchpadMK2Bridge.midi.xml: generated from the code mapping.
- tests/: knob engine, routing/LEDs/acceptance, emulator + live sender.
- docs/: DECISIONS.md, SETUP.md, EMULATOR.md, VERIFICATION.md.
- agents/: SPEC.md, PLAN.md, TASKS.md, ACCEPTANCE.md (build contract).